2017 EXAMINATIONS

Lancaster University

**Part I (First Year)**

**SCHOOL OF COMPUTING AND COMMUNICATIONS**

**SCC.130 Software Development (1 hour & 30 Minutes)**

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- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

## Question 1

**1.a** Consider the five-component framework of an information system:

i. List three elements that make up the hardware component and describe their functions.

**[3 marks]**

ii. List and briefly describe the functions of three elements as part of the software component that coordinate communications amongst the hardware components and the rest of the information system.

**[3 marks]**

iii. Identify two information systems where the first creates more work for people and the second creates less work. Explain how these two systems differ and the implications this has on efficiency.

**[4 marks]**

iv. Suppose an organisation would like to use its customer facing information system via the Web as part of its e-business strategy. Using an example, explain what type of architecture and associated functionality would the organisation need to consider. Further, discuss benefits and risks associated with such a strategy.

**[5 marks]**

**1.b** Mobile systems are considered to be information systems that support users in motion. In addition, users tend to move not only geographically, but also from device to device. Another reason of the importance of mobile systems is the size of their market and their popularity among consumers. As is shown in Figure 1 below, the major elements in a mobile system are user mobility, the device, connectivity and resources.

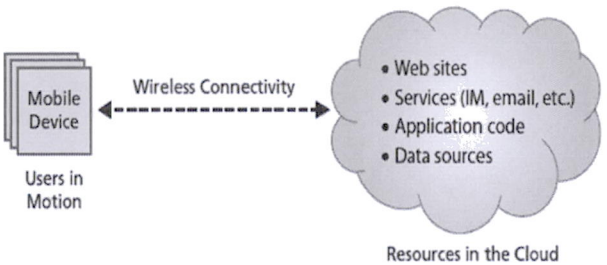


Figure 1: Elements of a mobile information system

**Question 1 continues on next page...**

**Question 1 continued.**

Using the five-component information system framework, identify and discuss the impact mobile-systems have for each of these five components:

i. Hardware **[2 marks]**
ii. Software **[2 marks]**
iii. Data **[2 marks]**
iv. Procedures **[2 marks]**
v. People **[2 marks]**

**[Total 25 marks]**

## QUESTION 2 (25 marks)

In this question, you will use the following three relations:

| films | | | |
|---|---|---|---|
| **fid** | **fname** | **boxoffice** | **genre** |
| 1 | Rocky | 225000000 | Sports Drama |
| 2 | The Godfather | 245100000 | Crime |
| 3 | Gone With the Wind | 390000000 | Romance |
| 4 | Forrest Gump | 677900000 | Comedy |

| actors | | |
|---|---|---|
| **aid** | **aname** | **age** |
| 1 | Sylvester Stallone | 70 |
| 2 | Talia Shire | 70 |
| 3 | Al Pacino | 76 |
| 4 | Tom Hanks | 60 |
| 5 | Robin Wright | 50 |

| performance | | |
|---|---|---|
| **aid** | **fid** | **year** |
| 1 | 1 | 1976 |
| 2 | 1 | 1976 |
| 3 | 2 | 1972 |
| 4 | 4 | 1994 |
| 5 | 4 | 1994 |

**2.a** i. Which three data types would be most suited for the 'actors' relation?

**[3 marks]**

ii. Define *two* types of relational data integrity.

**[4 marks]**

iii. List where the foreign keys are required in the above relations. What restrictions do foreign keys enforce on values?

**[3 marks]**

iv. When constructing this database in practise, it is common to implement a three-level ANSI-SPARC architecture. Define the three schema levels and explain what the schemas were designed to support.

**[5 marks]**

**Question 2 continues on next page...**

**Question 2 continued.**

**2.b** Write SQL statements that will provide answers to the following:

i. List all films whose box office is greater than $4 billion.

**[2 marks]**

ii. Alter the genre of the film 'Gone With the Wind' from 'Romance' to 'Drama Romance'.

**[3 marks]**

iii. List the film name, actor name, and the average box office revenue per year since film release. Only include films that have corresponding actors listed, and sort by the highest annual average box office.

**[5 marks]**

**[Total 25 marks]**

## QUESTION 3 (25 marks)

**3.a** Define what is meant by digital exclusion. Describe 2 causes of digital exclusion on society.

**[5 marks]**

**3.b** i. What legislation protects people's personal data?

**[2 marks]**

ii. Name and describe 4 of the 8 principles of this legislation?

**[8 marks]**

**3.c** Give three UK legislations for state surveillance. What do they cover?

**[6 marks]**

**3.d** Give 4 types of malware and explain their purposes.

**[4 marks]**

**[Total 25 marks]**

## Question 4

**4.a** Consider the link below between two systems across an inter-network.

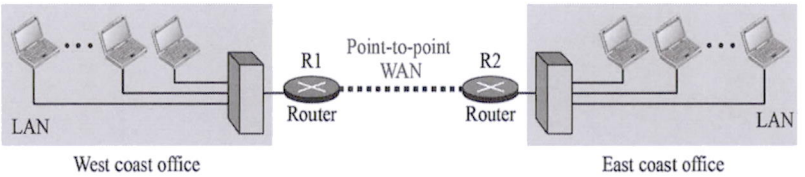


Figure 4.a: A point to point WAN link

i. Explain the concept of layering when communication takes place between the two systems.

**[2 marks]**

ii. Using the inter-network in the figure above, illustrate the layers of the TCP/IP protocol suite used and the flow of data when the two hosts, one from each side, exchange messages.

**[4 marks]**

iii. Considering that information systems rely on data communications and network systems in their operations, identify the three main criterias that need to be fulfilled for an effective and sustainable operation of the system. Explain, using an example, the importance of fulfilling this criteria.

**[6 marks]**

**4.b** This question relates to system analysis and design as part of the system development life cycle (SDLC):

i. Describe the different steps used in SDLC. In which step of the SDLC are the data flow diagram and decision tables' processes are used? Further give an example for each process: a data flow diagram and a decision table.

**[5 marks]**

ii. Using a real-life case study, discuss the challenges presented to an organisation when it considers its system's software conversion. Give your opinion, with justification, on what you think is the best adoption strategy for the system's software conversion.

**[4 marks]**

**Question 4 continues on next page...**

**Question 4 continued.**

iii. Disaster recovery planning is an important task for any organisation's information and software system. Discuss how a disaster recovery process could be implemented as part of a SDLC. Use an example of how this process can be implemented by taking into account system recovery and business continuity.

**[4 marks]**

**[Total 25 marks]**

**---End of Paper--**